

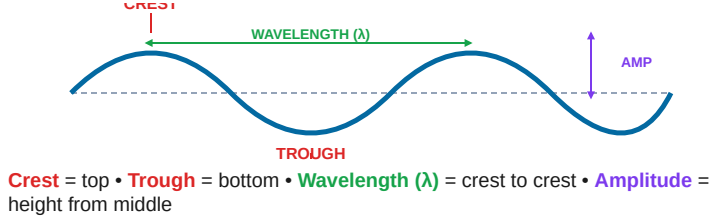
**The Big Idea** S8P4

A **wave** is a disturbance that transfers **energy** from one place to another **without transferring matter**. There are two main types of waves: **electromagnetic (light) waves** and **mechanical (sound, water) waves**. They behave very differently!

**Wave Parts: Transverse vs. Longitudinal** S8P4a**TRANSVERSE WAVES**

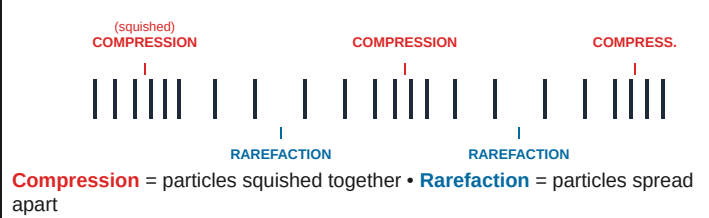
Particles move **perpendicular** (up & down) to wave direction.

Examples: Light, all EM waves, ocean waves, waves on a rope

**LONGITUDINAL WAVES**

Particles move **parallel** (back & forth) to wave direction.

Examples: Sound waves, slinky pushed forward

**Electromagnetic vs. Mechanical Waves** S8P4a

Feature	⚡ ELECTROMAGNETIC (EM)	🔊 MECHANICAL
Needs a medium?	NO — can travel through empty space!	YES — needs matter (solid, liquid, or gas)
Type of wave	Always TRANSVERSE	Can be transverse OR longitudinal
Speed in vacuum	Speed of light (~300,000 km/s) — FASTEST	0 — sound can't travel in space!
Examples	Visible light, radio, microwaves, X-rays	Sound, water waves, earthquakes
Speed is fastest in...	Vacuum/empty space (slows in dense matter)	Dense solids (slowest in gas like air)

GEORGIA CONNECTION Georgia is "Hollywood of the South" — the **film industry** (SS8E2c) records both kinds of waves on every set: **cameras capture light (EM)** and **microphones capture sound (mechanical)**. That's why a movie can show explosions in space, but the booms you hear are added later — sound truly cannot travel through a vacuum!

**The Electromagnetic Spectrum** S8P4b, S8P4c

All EM waves travel at the speed of light, but they differ in **wavelength**, **frequency**, and **energy**. As wavelength gets **SHORTER**, frequency gets **HIGHER** and energy gets **STRONGER**.

RADIO longest λ lowest E	MICRO-WAVE	INFRARED (heat)	VISIBLE LIGHT	UV	X-RAYS	GAMMA shortest λ highest E
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← LONGER wavelength, LOWER energy | SHORTER wavelength, HIGHER energy →

Memory trick: "Raging Martians Invaded Venus Using X-ray Guns"

**Practical Applications (S8P4c):**

Wave	Used For	Real Example
Radio	Communication	FM/AM radio, TV broadcasting, Wi-Fi, cell phones
Microwave	Cooking, communication, radar	Microwave ovens, satellites, radar (used by WWII Bell Bomber pilots)
Infrared	Heat, night vision	TV remotes, thermal cameras, heat lamps
Visible	Sight	The only EM waves your eyes detect (ROYGBIV)
UV	Sterilizing, tanning/burning	Sunlight, "black lights," sterilizing equipment
X-rays	Medical imaging, security	Doctor's office, airport scanners
Gamma	Cancer treatment, sterilizing	Hospitals, food sterilization (highest energy = most dangerous)

Flip over for wave behaviors, sound, and how density affects waves →



How Waves Behave: 5 Wave Behaviors S8P4d

When a wave hits a new material, one of these 5 things happens:



REFLECTION

Wave **bounces back**. Light off a mirror; **echo** = sound reflected.



REFRACTION

Wave **bends** when entering a new medium. Straw "bends" in water.



ABSORPTION

Material **takes in** the wave's energy. Black shirt absorbs light → gets warm.



DIFFRACTION

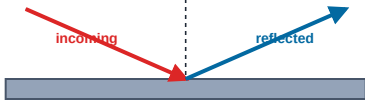
Wave **bends around** an obstacle or through an opening.



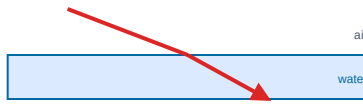
TRANSMISSION

Wave **passes through** a material. Light through glass.

Reflection (Echo)



Refraction (Bending)



How We See Color

Objects **absorb** some colors and **reflect** others. A red shirt looks red because it **reflects red** light and **absorbs** all other colors. White = reflects all; Black = absorbs all.

Density of Medium Affects Wave Speed S8P4e

The **density** of the medium changes how **FAST** a wave travels. **Sound and light behave OPPOSITELY!**



SOUND (mechanical)

Faster in DENSER mediums — particles are closer, so they pass vibrations along quicker.

Medium	Speed (approx.)
Solid (steel)	~5,000 m/s ⚡⚡⚡⚡
Liquid (water)	~1,500 m/s ⚡⚡
Gas (air)	~340 m/s ⚡
Vacuum (space)	0 (no sound!) ❌



LIGHT (electromagnetic)

Slower in DENSER mediums — atoms get in the way!

Medium	Speed (approx.)
Vacuum/space	~300,000 km/s ⚡⚡⚡⚡
Air	~299,700 km/s ⚡⚡
Water	~225,000 km/s ⚡
Glass	~200,000 km/s ⚡

Bonus: Waves in Georgia Studies SS connections

MLK's "I Have a Dream" Speech (SS8H11b)

Dr. Martin Luther King Jr. (born in Atlanta) and John Lewis spoke at the 1963 March on Washington. MLK's voice traveled to millions through wave conversions:

Sound (mech) → Mic → Electrical → EM (radio) → Speakers → Sound (mech)

CNN: 24-Hour News from Atlanta (SS8H12, SS8E2c)

Ted Turner founded CNN in Atlanta in 1980 — the first 24-hour news channel. CNN uses **radio waves**, **microwaves**, and **satellite signals** (all EM waves) to broadcast worldwide at the speed of light.

Bell Bomber Plant & WWII Radar (SS8H9b)

The Bell Bomber Plant in Marietta built B-29 bombers that used **RADAR** — a system that sends out **microwaves**, lets them **reflect** off enemy aircraft, and times the echo to find distance. Pure wave science applied to history!

Hollywood of the South (SS8E2c)

GA's film industry uses **cameras** (capture visible light/EM waves), **microphones** (capture sound/mechanical waves), and **green screens** (use light reflection). Marvel and Stranger Things filmed here!

ACTIVITY 1: Label the Waves

For each example, write **EM** or **MECH** and **T** (transverse) or **L** (longitudinal):

- Sunlight reaching Earth → _____
- Your friend talking to you → _____
- X-ray at the doctor → _____
- Wave traveling along a slinky pushed forward → _____
- Wi-Fi signal → _____
- An ocean wave at Tybee Island → _____

ACTIVITY 2: Identify the Behavior

Reflection / Refraction / Absorption / Diffraction / Transmission?

- Hearing your voice echo in a tunnel: _____
- A black T-shirt getting hot in the sun: _____
- Sunlight passing through a window: _____
- A pencil in a glass of water looks bent: _____
- Hearing a friend around the corner: _____

ACTIVITY 3: EM Spectrum Detective

- Which has **MORE** energy: a radio wave or a gamma ray? _____
- Which has a **LONGER** wavelength: visible light or microwaves? _____
- What type of EM wave does a TV remote use? _____
- Why can't sound travel through outer space, but light can? _____

I Can... (Self-Check for Milestones)

- ☐ Compare/contrast EM and mechanical waves (S8P4a)
- ☐ Identify transverse vs. longitudinal waves
- ☐ Label crest, trough, wavelength, amplitude, compression, rarefaction
- ☐ Order EM spectrum from low → high energy (S8P4b)
- ☐ Match EM waves to real-world uses (S8P4c)
- ☐ Define reflection, refraction, absorption, diffraction, transmission (S8P4d)
- ☐ Explain why we see color
- ☐ Predict how density affects sound vs. light speed (S8P4e)
- ☐ Connect waves to Georgia history (MLK, CNN, Bell Bomber, film)